

Volume Effect on Return and Volatility Spillover in Dually Listed Securities

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Abstract

We empirically investigate the volume impact on serial correlations in return and volatility of 168 ADRs listed in the NYSE market and their underlying securities from 26 developed and emerging markets in the world. Using a VAR setup, we forecast volumes of cross listed securities accounting for their comovements and denote the residuals as unanticipated volumes. We estimate return and volatility spillover, serial auto and cross correlations between ADRs and their underlying home shares and find asymmetries in those spillover. Further, using two interaction terms, unanticipated volume times lag return for each ADR and its underlying security, we determine volume surprise induced spillover and find those spillovers are also asymmetric between ADRs and their underlying home securities and between serial auto and cross correlations. Finally, we classify the data based on positive, negative, or zero momentum and find that classification explains asymmetry in how firm size and liquidity determine continuation or reversal. We also find momentum as a factor in explaining asymmetry in volatility spillover. Our evidence suggests that volume residuals induced return and volatility spillover denote multiple trading motives including information and liquidity, but perhaps most strongly momentum revealed in trades. Many of our findings either contradict existing results or are new to the literature on cross listed securities.

1. Introduction

Using a panel of daily data on 168 ADRs listed on the New York Stock Exchange (NYSE later) and their underlying securities from 26 developed and emerging markets around the world over a period of 8-22 years, we empirically determine the impact of volume on the return and volatility dynamics of cross listed securities. Specifically, we acknowledge the evidence in Halling et al. (2009) that trading volumes of cross listed securities may commove with their own and cross lagged volumes and thereby address the following questions: How does volume surprise or unanticipated volume adjusted for comovements in volumes between pairs of cross listings impact return and volatility

spillover, serial auto and cross correlations of and between ADRs and their underlying home securities? Further, are there common market/country and security specific determinants of volume surprise induced return and volatility spillover, which may indicate how information content in volume is incorporated into return and volatility dynamics?¹

Two existing strands of empirical research on return and volatility spillover provide the foundation of our investigation. First, Conrad et al. (1994) observe in a sample of NASDAQ stocks that the volume impact on first order autocorrelation in return or return spillover in individual stocks is asymmetric such that low (high) trading volume stocks experience return continuation (reversal). Llorente et al. (2002) broaden the sample to include all US stocks listed on the NYSE and the AMEX and explain the evidence on continuation vs. reversal in volume induced return spillover as due to the higher likelihood of information vs. hedging or allocation motivated trades as implied by Campbell et al. (1993) and observed in small vs. large cap stocks respectively. Gagnon and Karolyi (2009) extend this literature to include cross listed securities and document that firms with low (high) levels of information asymmetry witness return reversals (weaker reversal or continuation) in one market following high volume in the alternate market; in addition, they report that return spillover, be it continuation or reversal, is more pronounced when it originates from the home market rather than the host (US) market.

¹ We use the term spillover rather broadly to indicate serial autocorrelations for a security and also serial cross correlations between matching pairs of securities. Note that while serial correlations in returns are mentioned in the literature as spillover, serial autocorrelations in volatility are mostly referred to as persistence and only serial cross correlations as volatility spillover in cross market analysis. We also use the terms volume surprise and unanticipated volume interchangeably throughout the paper. In this paper, our primary interest is to analyze the volume impact on serial cross correlations in returns and volatility persistence.

Second, with respect to the impact of trades on asymmetric time varying serial autocorrelations in volatility of domestic US stocks, Abramov et al. (2006) distinguish between informed (contrarian) and uninformed (herding or liquidity) trades and find empirical support for the seemingly counterintuitive implications in Wang (1993, 1994) and Campbell et al. (1993) that volatility must decrease (increase) with information (hedging or liquidity) motivated trading.² However, it's not known how trades may impact the volatility dynamics in cross listed securities.

The existing literature on the information content of unanticipated volume on return and volatility spillover with respect to cross listed securities fails to consider volume comovements between corresponding pairs of cross listed securities for the purpose of computing expected volume for a security, which casts doubt on the reported results of volume impact on return spillover in, for example, Gagnon and Karolyi (2009), who deseasonalize volume denoted by the number of trades for cross listed securities with a 50 days MA filter for the purpose of determining unanticipated excess (+/-) volume. Further, the approach in Gagnon and Karolyi (2009) to identify a set of cross sectional firm and country specific determinants of return spillover is unconvincing since many of those cross sectional measures are time invariant and hence do not adequately capture the time dynamics in trading and prices in a diverse sample of cross listed securities. Next, while Abramov et al. (2006) provide evidence of trades causing asymmetric volatility in domestic US shares, no empirical evidence exists regarding how volume surprise may cause asymmetric volatility spillover in and

² Lo and Mackinlay (1990) and Sentana and Wadhwani (1992) propose two distinct market microstructure models based on asynchronous vs. feedback trading to justify volatility changes in equity markets.

between cross listed securities. Finally, despite the lack of empirical evidence, it stands to reason that return and volatility spillovers have a common set of time varying cross sectional determinants.

Empirically, we address these concerns as follows. First, we follow Halling et al. (2013), who test the predictions of theoretical models of multi market trading volume by Chowdhury and Nanda (1991), Pagano (1989), and Menkveld (2008) and run a bivariate VAR model incorporating cross security volume for estimating trading volume forecasts of each pair of ADR and its underlying home security, in which the prediction errors denote volume surprises. Second, we estimate the auto and cross serial correlation coefficients along with those related to two interaction variables denoting volume induced lag returns, parameters of another bivariate VAR model for ADR and home security returns and revisit Gagnon and Karolyi (2009) to test the implications of Campbell et al. (1993), Wang (1994), and Blume et al. (1994). We propose a momentum indicator based jointly on trade and return to capture a non-stationary return volume dynamics. Third, we extend Abramov et al. (2006) and test the implications of Wang (1993, 1994) in terms of volume impact on volatility spillover for a sample of cross listed securities. We denote volatility as absolute error of returns and estimate the parameters of a bivariate VAR model for volatility in which not only lagged variance but lagged cross covariance, and volume surprises induced volatility enter into each security's variance function. Finally, in a regression setup, we determine the effect of size, liquidity, market type (developed and emerging), and time overlap (large, small, and no) on the return and volatility spillover parameter estimates for different

subsamples of momentum.

Our empirical results find the following. First, there is significant evidence of short term memory in volume and volume comovements, albeit asymmetric, across a majority of ADRs but all of their corresponding home shares. Second, first order serial return autocorrelations are polar opposites- negative for ADRs and positive for their underlying home shares, while return cross correlations are uniformly positive for both ADRs and the corresponding home shares. Although the number of significant coefficients associated with the volume induced correlations are somewhat limited, the mean autocorrelations are generally positive for both ADRs and their underlying home shares indicating continuation, while the mean volume induced cross correlations are positive implying continuation for ADRs but negative denoting reversal for their underlying home shares. Third, tests of volatility spillover indicate significant auto and cross correlations in volatility for both ADRs and their underlying securities supporting bi-directional causality. However, while volume surprise do not appear to reduce volatility persistence of either ADRs or their underlying home securities, volatility spillovers including volume surprise induced spillovers indicate both continuation and reversals, albeit heavy on the reversal side, for both ADRs and their underlying home shares. Fourth, we find security level differences in size and/or whether securities are, historically inclined to continuation or reversal cast more impact on return and volatility spillover than country/market level differences. We interpret our evidence as indicative of volume surprise induced return and volatility spillover originating from multiple trading motives including information but first and foremost, liquidity.

Our research contributes to the literature in multiple ways. First, we follow Halling et al. (2013) and decompose volume into expected volume and volume surprise based on multiple market trading in which the lagged trading volumes of both ADRs and their underlying home securities are among the determinants of each security's trading volume. Thereafter, we revisit the evidence in Gagnon and Karolyi (2009) regarding unanticipated volume effect on return spillover from a comprehensive set of dually listed securities from multiple developed and emerging markets. Our results contradict many of the results in Gagnon and Karolyi (2009), particularly those related to volume induced return spillover. Second, in our knowledge, this is the first study of its kind that examines unanticipated volume effect on volatility spillover for a comprehensive set of dually listed securities from multiple developed and emerging markets. Third, through a robust regression methodology, we determine common security and market/country specific factors which explain variations in the volume induced auto and cross correlation coefficients in return and volatility. Results from our study show short term predictability in return and volatility for cross listed securities and hence refute security level cross market integration, which in turn opens up potential for risk arbitrage that has direct practical ramifications for international portfolio management, especially in tactical asset allocation and hedging.

2. Literature Review

We document the effect of unanticipated trading volume or volume surprise on return and volatility spillover, serial auto and cross correlations for cross listed securities in

home and host (NYSE) markets. In addition, we investigate whether there exist security and/or market/country level determinants of return and volatility spillover. This research question overlaps diverse areas of the existing finance literature and hence for the purpose of exposition, we break down the literature as follows: a) Volume surprise model for cross listed securities b) Return and volatility spillover and further unanticipated volume effect on such spillover c) Security and country/market specific determinants of return and volatility spillovers. We provide a brief survey of the existing literature often overlapping two or more topics mentioned above.

Lo and Wang (2010) summarize trading volume definitions used in the literature along with some of their distributional properties. Trading volume of single listed shares may contain a unit root, a deterministic or stochastic trend, a long memory component, or even a combination depending on a variety of factors including estimation intervals and sample period (Lo and Wang [2000], Fleming and Kirby [2009], Chae [2005]). Brock and Kleidon (1992), Huberman (), and Brown et al. (2009) contend trading volume may contain multiple motives including information, inventory, and momentum.

For cross listed shares, Pagano (1989), Chowdhury and Nanda (1991), and Menkveld (2008) propose theoretical models of multimarket trading volume with discretionary traders and conclude that any optimal trading rule includes the possibility that all or a substantial portion of trading is concentrated in only one market. Empirically, Halling et al. (2013) find continued dual market trading for cross listed shares and that while there is significant correlation between trading

volumes across multiple trading venues, the breakdown of trading volume among those alternative venues is not consistent among firms and markets and further two time varying factors, legal enforcement regime (market/country specific) and institutional ownership (security specific) impact the volume of discretionary trading in each market.

Regarding trading volume impact on autocorrelation and/or cross correlation between returns on domestic securities, Conrad et al. (1994) explore the nexus between trading activity and short horizon (weekly) return comovements for NASDAQ listed securities and report that high (low) activity stocks experience return reversals (continuation of positive correlation); Llorente et al. (2002) deduce empirical implications from Campbell et al. (1993) and Wang (1994) and explain the return correlation continuation (reversals) for low (high) activity stocks as due to higher (lower) level of information asymmetry and informed trading in small cap stocks, which also exhibit high bid ask spread.

Gagnon and Karolyi (2009) extend the dynamic volume return literature by adding cross listed securities in the mix. In a comprehensive study of 556 dual listings, US and their underlying home securities, they find that firms with low (high) levels of information asymmetry witness return reversals (weaker or continuation) in one market following high volume in the alternate market; in addition, they report that return spillover, be it continuation or reversal, is more pronounced when it originates from the home market rather than the host (US) market. Empirically, Gagnon and Karolyi (2009) use a two-step process. First, they estimate the autocorrelation and

cross correlation parameters along with their interactions with security specific trading volumes from a time series regression model for each security pair over each quarter; and then in a pooled cross sectional regression model in which the volume induced spillover parameters are the dependent variables, they estimate the coefficients associated with several information asymmetry measures as independent variables.

In terms of volume volatility relation, multiple empirical issues have emerged including those related to spillover and role of volume on volatility persistence in cross listed securities.³ First, although a long list of studies have investigated whether for broad markets volume and volatility are contemporaneously correlated (MDH) or alternately there exists a bi-directional lead lag relation (SIAH) between the two, only a few studies report on similar volume volatility nexus for individual securities. Bollerslev and Jubinski (1999) find volume and volatility of US stocks are fractionally integrated and share a common long memory parameter; however, Fleming and Kirby (2007) report that 20% of their sample of 20 middle market securities do not have identical long memory in volume and volatility and Anè and

³ Karpoff (1987) surveys the literature on the relation between changes in trading volume and absolute price changes. Two contrasting hypotheses, the Mixture of Distribution (MDH) and the Sequential Information Arrival (SIAH) hypotheses exist to explain the volume volatility relations. Harris (1986), Smirlock and Starks (1986), Lamoureux and Lastrapes (1990), Kim and Kon (1994), Anderson (1996), and Gallo and Pacini (2000) find evidence in support of a contemporaneous volume volatility relation in the US stock market, as predicted by the MDH. Chen and Tse (1993), Omran and McKenzie (2000), Pyun et al. (2000), and Bohl and Henke (2003) report similar evidence from the Japanese, UK, Spanish, Korean, and Polish stock markets respectively. In contrast, the SIAH (Copeland [1976]) predicts bidirectional causality between volume and volatility. Saatcioglu and Starks (1998) find positive correlation between changes in trading volume and price changes but mixed evidence on the direction of causality in Latin American markets. Campbell et al. (1993), Hiemstra and Jones (1995), and Brooks (1998) report bidirectional Granger causality between daily stock returns and changes in trading volume. Gallant et al. (1992) and Silvapulle and Choi (1999) find one-way Granger causality from volume to volatility in US and Korean stock markets respectively but Lee and Rui (2000) and Lee and Rui (2002) do not find trading volume Granger cause returns in Chinese and Japanese markets respectively. Yet, Chan et al. (1996), Jones et al. (1999), Giot (2010) motivate another strand of the empirical literature on volume volatility comovements and question if trading volume measured as such is indeed the driver of the volume volatility relation for individual securities.

Ureche-Rangau (2008) find that volume and volatility share only short term comovement but no long term memory. On the other hand, Darrat et al. (2003) report one-way Granger causality from trading volume to return volatility for 24 out of 30 DJIA stocks. Li and Wu (2006) reports non-linear correlation between volume and volatility of a sample of US stocks.

A second question even more pertinent to this research is how volume changes the serial auto and cross correlation coefficients associated with volatility of cross listed securities. This question nests three separate queries- How volatility transmits across securities in multiple markets? Whether and if so how volume alters volatility persistence in individual securities and markets? And finally, how volume impacts volatility persistence and transmission across cross listed securities?

Gagnon and Karolyi (2006) review the literature on volatility spillover across geographically dispersed markets and report inconclusive evidence with respect to the volatility transmission across country indexes. From a study of multiple developed and emerging markets, Diebold and Yilmaz (2009) observe fundamental differences in stochastic properties between return and volatility spillover- return spillover shows trend but no busts while volatility shows no trend and only busts.

The issue of volume effect on volatility persistence has been studied, albeit only cursorily and in the context of market indexes, by Chen et al. (2001), Girard and Biswas (2007), and Vlastakis and Markellos (2012). Those studies report that the coefficient associated with lag volatility persists even after adding volume into a volatility model.

On the other hand, while evidence of serial autocorrelations in volatility or volatility clustering abounds in the finance literature, the empirical evidence with respect to cross security volatility transmission from ADR (host) to its underlying (home) and reverse for dually listed securities is limited, mostly country specific, and scattered.⁴ Chan et al. (1996) compare the intraday volume, volatility, and spread of Japanese and European ADRs with those of the underlying securities and Hauser et al. (1998) investigate the information flow between NYSE and Tel Aviv exchange for five stocks listed in both exchanges. Yang (2007) and Chen et al. (2010) model return and volatility spillover for Japanese and Chinese ADRs respectively. Following up on Karolyi (1995), Xu and Fung (2002) and Poshakwale and Aquino (2008) fit bivariate GARCH models for Chinese ADRs, and a global basket of ADRs and their underlying stocks respectively where only the inter security volatility functions are interrelated but cross security volumes are not integrated into the volatility models.

3. Institutional Structure of NYSE Foreign Listings, Data, and Preliminary Results

3.1 Data description and Institutional structure of ADR Listings

As of December 31, 2012, 525 common and preferred equities of foreign corporations from 46 countries are listed on the NYSE and the NYSE Market among which 259 trade as ADRs (including Global Shares and NY Registry Shares) with primary listing

⁴ Refer to the literature surveys on GARCH regarding volatility clustering and persistence. Much of the empirical literature on intercountry return and volatility transmission is based mostly on country indexes. For a literature review, refer to Karolyi (2006). Recently, Chen et al. (2009) report on volatility transmission on country indexes denoted by respective ETFs. Diebold and Yilmaz (2009) and Bekaert et al. (2009, 2014) study volatility transmission within and across markets to test time varying financial market integration and spread of contagion across markets. We focus on inter security volatility transmission, instead.

in a domestic exchange located in a foreign country.⁵ Table 1 provides a breakdown of a sample of 168 ADRs and their corresponding home country shares listed in 29 foreign exchanges located in 17 developed and 12 emerging markets (MSCI Developed and Emerging Markets Indexes include 16 and 11 of those markets) respectively.⁶

We briefly discuss below the institutional details of the ADR listing process at the NYSE and the supporting ADR database to motivate the empirical framework adopted in this paper. A foreign corporation may choose to list on the NYSE, the main exchange for established mid to large cap companies or the NYSE Market for relatively new and smaller companies, both of which currently (since 2012) support a designated market maker (DMM) for each listing along with access to Supplementary Liquidity Providers (SLP) usually for highly liquid securities. The NYSE and NYSE Market listing allows foreign corporations to market their securities to US investors in the form of depositary receipts known as ADRs, which are sponsored by one of the designated US depositary banks; simultaneously, it allows US investors to directly invest in shares of foreign corporations. Prior listing in its home or another foreign jurisdiction is not mandatory for a foreign issuer and hence while at the beginning of the ADR program, only large

⁵ Among the 525 NYSE and NYSE MKT listed ADRs from 46 countries, a majority of recently listed ADRs from China and also a few other emerging markets listed on NYSE MKT do not have a primary listing. All issuers from Canada and a few small nations/territories along with a majority of issuers from China have issued original listings (ORD) instead of ADRs (ORDs do not require sponsorship by a depositary bank- for more details, please refer to <https://research.scottrade.com/knowledgecenter/Public/help/Article?docId=37010be1721740e0879fb4b3510db8ed>); incidentally, those are also the two countries with the highest number of foreign issuers at the NYSE market. Global and NY registry shares are for securities from Netherlands and Luxembourg, which prohibit domestic companies from selling shares in another currency.

⁶ Our primary criterion for identifying developed vs. emerging economies is based on MSCI ACWI indexes of 23 Developed and 23 Emerging Markets as of July 2014 as listed in Table 1. As per MSCI ACWI components list, eight developed markets (Austria, Canada, Hong Kong, New Zealand, Singapore, Sweden) and seven emerging markets (Czech Republic, Egypt, Hungary, Malaysia, Qatar, Thailand, and UAE) do not have any ADR listed on the NYSE Market. Further 23 ADRs/home country shares are either preferred shares and/or do not have the necessary data and hence are excluded. Included in those 23 securities are two each from Colombia and Russia and one each from Peru and Turkey, which denote their entire ADR listings respectively and are therefore completely eliminated from the sample due to missing data. Hence our final sample consists of 142 securities from 26 developed and emerging markets.

and reputable foreign companies would list and raise capital in the US, since the mid-2000, many small unlisted foreign companies have bypassed their domestic capital markets and resorted to simultaneously issuing IPOs and ADRs in the USA; a majority of those foreign issuers have subsequently raised capital and listed on a primary exchange in their respective home countries.

3.2 Descriptive statistics on ADR and their corresponding home security volume and return

We obtain daily data on closing prices, number of shares traded, and the value of traded shares (turnover) for those 168 ADRs and their underlying home securities listed on the respective domestic Stock Exchanges from Bloomberg data terminals for a period of approximately 8-22 years based on availability, from the initial trading day for each security beginning on or after April 1983. Hence, the sample period for each pair of ADR and its underlying security begins from their respective initial trading dates from/after April 1, 1983 and ends on December 31, 2013. We also obtain similar data on prices and volume for NYSE and for those 26 developed and emerging markets (composite market indexes, wherever available), where our sample securities have their primary listings along with the USD to home currency exchange rates for all the countries in our sample.

We denote volume as $v_t^i = \ln(N_t^i)$ and compute returns as $r_t^i = \ln\left(\frac{P_t^i}{P_{t-1}^i}\right)$ where $N(i, t)$ and $P(i, t)$ denote number of trades and closing price respectively of the i th security on the t th day. Table 2 provides descriptive statistics on trading volume and daily returns for our sample ADRs and their underlying securities listed on home

exchanges. It also provides country wise summaries of those statistics. The descriptive statistics indicate large differences between means of daily returns for both ADRs and their corresponding home securities, albeit some of the differences seem to be due to the varying length of each security's time series. In fact, while most ADRs show small positive mean daily returns, most home securities show negative often large mean daily returns during the sample period. A few exceptions are ALU and TOT (France), AEG, ING, PHG (all three securities in The Netherlands), and three seemingly outlier securities, BBVA, SAN, TEF (Spain), which exhibit large positive home country returns during the sample period. While standard deviations of ADRs and their underlying home shares returns look similar, there are visible differences among the mean returns and standard deviations on a country by country basis.

In contrast, for a majority of securities, trading volumes in home shares are conspicuously larger than those in ADRs, for example and although not a common occurrence by any means, BCA's (Chile) average trading volume of home shares is 8572 times its average volume at NYSE. However, there are 23 securities including VIV and VALE (Brazil), ELN (Ireland), and TEVA (Israel) for which the ADR volumes are higher than those of home shares. Among the sample countries, only Brazil and Israel show average trading volumes in ADRs higher than that of home shares. Japan (345 times) and China/Hong Kong (198 times) top the list of all the remaining countries whose average trading volumes in home exceeds that in ADRs.

Figure 1 graphs the risk return profile i.e., the distribution of risk denoted by standard deviations against means of daily return for ADRs and their underlying home

shares, which seem to indicate almost similar risk return profile for ADRs and their corresponding home shares. Since each pair of ADR and its underlying cannot be visually tracked on the left and right graphs, this may be misleading and hence we conduct formal tests of joint mean variance equality for our sample ADRs and their corresponding home shares. In spite of the seemingly similar distributions of returns for ADRs and their underlying home shares, a joint means and variances test for the equality of distributions between daily returns reject the null hypothesis for a large number of ADRs and their corresponding home securities.⁷

3.3. Trading volume expectation model and volume surprise

We denote volume by the total number of shares traded and decompose it into expected and unanticipated (surprise) components as follows.⁸ For each pair of ADR and its underlying home share in our sample, we use a modified version of Halling et al. (2013) with the US and the respective home market volumes as control variables as in 1a-1b below to forecast trading volume based on a VAR methodology with the corresponding forecast error term, $V_{i,t}$, as the volume surprise.⁹ We concur with the existing literature and focus on the information content in $V_{i,t}$, the unanticipated component of volume or volume surprise.

⁷ We do not report the joint mean variance test results. We also find significant differences between the distributional properties of day and daily returns, which seem to indicate bias in the results of those studies which combine day and daily returns to account for large time differences between markets, which do not have overlapping market open hours. He and Yang (2012) report differences between daytime and overnight returns spillover for Chinese ADRs and their home shares.

⁸ We also use turnover as a measure of volume. The distributional properties of turnover are similar to those of market value of equity and significantly differ from those of number of trades.

⁹ We recognize that market volume creates asymmetric effects between active and thinly traded stocks. Liquid stocks may suffer a larger impact of market volume changes than their thinly traded counterparts. This asymmetry may spill over to home and/or host markets in a largely unpredictable way since while ADR volumes are customarily low in the US, a majority of ADR issuers are among the most active stocks in their respective home markets.

$$v_{i,t}^{us} = \alpha_1 + \beta_{11}v_{i,t-1}^{us} + \beta_{12}v_{i,t-1}^h + \phi_1v_t^{us} + V_{i,t}^{us} \dots (1a)$$

$$v_{i,t}^h = \alpha_2 + \beta_{21}v_{i,t-1}^h + \beta_{22}v_{i,t-1}^{us} + \phi_2v_t^h + V_{i,t}^h \dots (1b)$$

Table 3 contains country wise summaries of the parameter estimates from models 1a and 1b, and unit roots test (ADF) statistics for the residual, $V_{i,t}$ in equations 1a and 1b for our sample securities. The VAR parameters are OLS estimates with Newey-West correction for autocorrelation and heteroscedasticity. The t-statistics corresponding to the parameters and the model fit statistics given by the adjusted R^2 provide mixed evidence of volume comovements for ADRs and their underlying home shares. Adjusted R^2 for ADRs range from 16 to 22 percent, while those for home shares span between 16 and 73 percent.

All ADRs and their underlying home shares show consistently significant (at less than 1%), negative 1st order auto correlation; however, with regards to cross correlation, ADRs and home shares show evidence of sharply different levels of comovements. In terms of country wise distributions, cross security volume comovements are significant for 66 ADRs from 10 emerging and 14 developed markets; in contrast, 125 ADRs from all emerging and developed markets show significant cross market volume comovements. Among the ADRs and home shares, there are also differences in signs for the mean parameters- 11 countries show a negative lag home country volume impact on ADR volume, whereas all countries show a positive lag ADR volume impact on home share volume. This evidence of asymmetric, one sided effect, US to home but not the converse, is observed in cross country returns but given in terms of volume, ADR (US market) volume accounts for a smaller proportion of volume than that

accruing from the home market, is surprising nonetheless. We also note that the home securities from the Australia-Asia region, namely Australia (5), Hong Kong (14), India (6), S. Korea (7), Taiwan (7), and Japan (15) bear the largest number of significant cross correlations followed by Europe with 27 and the Americas with 17 home shares. We interpret this asymmetry as partially due to the extent of overlapping trading time zones between the rest of the world and the US. We will explore this issue of time overlap later in a substantive way.

Finally, almost all ADRs and their underlying home shares exhibit significant positive buoyancy from respective local market volume (not reported in Table 3); only six securities (2 each from Chile and Mexico, 1 each from the Netherlands and Spain) find insignificant local market volume effect. The four cited countries share more trading time overlap with the US than do countries in Australia and Asia.

The residuals from each of the equations show unanticipated volume or volume surprise and how it affects returns and volatility spillover is the fundamental issue we investigate in this paper. The residuals, $V_{i,t}$ for each ADR and its underlying home security are tested for autocorrelations via ADF ($\chi^2(6)$) and KPSS tests, which soundly reject unit roots for all ADRs and their underlying home shares.

4. Testing hypotheses on the effect volume surprise on return spillover

4.1 Effect of volume surprise on return spillover

We estimate the return serial auto and cross correlation parameters including those

induced by volume surprises as in Gagnon and Karolyi (2009) from the following set of bivariate VAR models, where $r_{i,t}^{us}$ and $r_{i,t}^h$ denote daily returns while $V_{i,t}^{us}$ and $V_{i,t}^h$ denote unexpected volume for the i th ADR or home security respectively, at time t . We include two interaction terms, lag unexpected volume * lag return corresponding to each ADR and its underlying home security, which denote the respective volume induced lag return comovements. We call these volume induced spillover. Finally, we include two relevant control variables, exchange rate return and contemporaneous market (NYSE composite for ADR and the respective home market corresponding to the respective underlying shares) return in each equation in the system. The exogenous control variables account for broad systematic risks borne by the securities in the given market. The resulting models are as follows.

$$r_{i,t}^{us} = a_1 + b_{11}r_{i,t-1}^{us} + b_{12}r_{i,t-1}^{us}V_{i,t-1}^{us} + b_{13}r_{i,t-1}^h + b_{14}r_{i,t-1}^hV_{i,t-1}^h + c_{15}x_t^{us} + e_{i,t}^{us} \dots (2a)$$

$$r_{i,t}^h = a_2 + b_{21}r_{i,t-1}^h + b_{22}r_{i,t-1}^hV_{i,t-1}^h + b_{23}r_{i,t-1}^{us} + b_{24}r_{i,t-1}^{us}V_{i,t-1}^{us} + c_{25}x_t^h + e_{i,t}^h \dots (2b)$$

The parameter vectors b_{ij} where $i=\{1,2\}$ and $j=\{1..4\}$ in equations 2a-2b signify time dependence and predictability of returns via serial auto and cross correlations and further those with lag volume interactions imply volume induced return spillover for each ADR and home security respectively. With respect to the above one lag auto and cross correlations parameter estimates, b_{ij} Blume et al. (1994), Wang (1993), and Campbell et al. (1993) stipulate the following alternate hypotheses:

- a) Return lag autocorrelation, $b_{i1} \geq 0$
- b) (Volume residual * Return) lag autocorrelation, $b_{i2} > 0$, for information motivated trades; (Volume residual * Return) lag auto correlation, $b_{i2} < 0$, for allocation/hedging motivated trades
- c) Return lag cross correlation, $b_{i3} \neq 0$

- d) (Volume residual * Return) lag cross correlation, $b_{i4} > 0$ for + volume residual;
 $b_{i4} < 0$ for - volume residual.

For each ADR and its underlying home share, we estimate the parameters of the above VAR model and report for each country summaries of parameter estimates and their corresponding statistical performance in Table 4A. The top rows of Table 4A contains the mean parameter estimates along with the corresponding number of positive and negative significant coefficients and the mean adjusted R^2 denoting model goodness of fit for each sample country. The bottom two rows contain summary statistics related to two variations of the model in equations 2a-2b. Between those bottom two rows, the uppermost row contains the VAR summary statistics for the exact model above; in the row below, we report, for comparison purpose, aggregate VAR estimates of the parameters of equations 2a-2b, where unanticipated volume is computed using a 50 days MA filter as in Gagnon and Karolyi (2009).

Parameter estimates for individual securities reported under each sample country indicate significant but asymmetric lagged dependence on returns for a majority of ADRs and their underlying home securities. The summary row finds 108 ADRs (129 home securities) display negative (positive) lagged dependence with barely a handful, 4 (5) ADRs (home securities) showing positive (negative) lagged dependence. This is interesting given the wide range of countries represented in the sample, a large majority of which show similar positive serial autocorrelations contradicting the evidence in Gagnon and Karolyi (2009) but confirming similar empirical evidence in Eun and Sabherwal (2003) and Grammig et al. (2005). By contrast, the negative autocorrelations for the ADRs confirm similar findings in Gagnon and Karolyi

(2009). These contrary evidence supports the implications of models of heterogeneous traders by Blume et al. (1994) and Wang (1994).

However, with respect to serial cross correlations in returns, only 24 (57) home securities and 23 (23) ADRs display significant positive (negative) lag cross dependence respectively. In this instance, although the evidence is weak in terms of number of significant cross correlation coefficients, our findings of a fairly high proportion of negative lag cross correlations for both ADRs and home securities contradict Gagnon and Karolyi (2009) and also He and Yang (2012) specifically for Hong Kong based ADRs. The evidence casts doubt on trades being ‘predominantly’ information based and support multiple trading motives implicit in trading (Brown et al. [2009]).

The lagged dependence in return, when volume innovation is included, irrespective of whether it is auto or cross dependence, shows a completely different picture with 64 (16) ADRs and 124 (11) underlying securities displaying significant positive (negatives) volume induced autocorrelation, while 43 (16) ADRs and 25 (51) home securities exhibit positive (negative) volume induced cross correlations. These results stand in stark contrast with those in Gagnon and Karolyi (2009), who find volume induced autocorrelation ‘reliably’ negative for both ADRs and their underlying home securities. Our evidence offers varying and mixed levels of support for Blume et al. (1994) and Campbell et al. (1993). While volume induced autocorrelation indicates strong support for information based trades at home, all other spillover parameters indicate mixed information and hedging motives for both

ADR and home securities trades; the high number of insignificant coefficients weakens the predictions of those models with heterogeneous traders. While in terms of number of securities with significant estimates, the evidence is quite weak, nonetheless, the mean volume induced cross correlation coefficient of -0.004 supported by 51 significantly negative parameter estimates for home securities confirms similar findings by Gagnon and Karolyi (2009).

Our model outperforms Model GK based on Gagnon and Karolyi (2009) in which unanticipated volume is measured as the excess over 50 days moving average. In general, while the summary statistics from the referenced models find a lot of consistency in the parameter estimates and the ratio between the number of positive and negative significant estimates; however, for the two volume induced cross correlation parameters, e_1 and e_2 , the signs alternate. Note that in both these cases there are large numbers of insignificant coefficients and further that the variability determined by the fraction of positive to negative coefficients is lower with Model 1. The positive (negative) auto and cross correlations denoting return continuation (reversal) completely refute similar evidence in Gagnon and Karolyi (2009) but supports mixed information and hedging/liquidity motivated volume induced spillover.

In table 4B, we report summary statistics on the parameters of equations 2a and 2b for two relevant subsamples, namely ADRs and their corresponding home securities from developed and emerging markets and markets with substantial, some, and no time overlaps with the US. The former is motivated by elevated trading and

market frictions in emerging markets compared to those in developed markets, which are documented to have caused differences between trading performances of ADRs and their underlying home shares. With respect to time overlap, we find some evidence of it being a factor in volume commovement that largely confers with similar findings in Halling et al. (2013) and hence we include three separate time zones with differing trading time overlaps with the US. We also provide results from a significance test for differences in means between those subsamples.

The subgroup results based on cross sectional classification of home shares in emerging vs. developed markets and in separate time zones, which proxy for emerging vs. developed markets and beyond indicate only a few scattered significant group mean differences. For example, the group mean parameters associated with lag cross return*volume for ADRs from developed (emerging) markets are 0.000 (0.029), significant at 10% level, while those for home shares are 0.029 (0.023), which are not significantly different. The respective group mean parameters associated with auto and cross lag returns for home shares from developed (emerging) markets are -0.151 (-0.030) and 0.129 (0.364), both significant at 1% level. Similarly, significant differences exist between respective group mean parameters associated with select auto and cross lag returns and lag return*volume for ADRs and their underlying home shares from high overlap, low overlap, and no overlap countries. The lack of significant results on specific country/market indicator variables confirm similar findings by Gagnon and Karolyi (2009).

We obtain little insight into the asymmetric volume impact on return spillover for and across ADRs and their underlying home securities from the above cross sectional classifications and argue that any cross sectional classification is perhaps inadequate for capturing return dynamics. Hence we devise a time series classification of the sample based on momentum trading, a return chasing trading strategy where excess volume is caused by high (low) trading when returns are higher (lower) than expected (also positive (negative) feedback trading), which we contend may explain the continuation reversal dynamics in return. Brown et al. (2010) indicate momentum content in trading volume.

We capture this momentum trading strategy with the help of a trinary dummy variable to separate volume residual*return into three momentum groups- a) positive momentum, M+: positive volume shock with positive return or positive feedback b) negative momentum, M-: negative volume shock with negative return or negative feedback c) zero momentum, M0: positive (negative) volume shock with negative (positive) return. We classify each firm trading day into positive, negative, and zero momentum days and re-estimate the parameters of the model in equations 2a and 2b. Since the data now do not retain the time series continuity, we estimate the OLS parameters of two separate single equations.

We expect that a breakdown of the sample by momentum classes may allow cross sectional differences to be more impactful during one momentum class or another. Table 4B Panel C reports summary statistics and results on the differences in means tests for each subsample classified as M+, M-, and M0. The results are striking.

Other than the parameters associated with auto lag return for ADRs and cross lag volume*return for home shares every pairwise comparison shows significant differences between the parameter estimates.

In the next section, using a regression analysis, we identify the determinants of the volume induced return spillover parameters for each momentum class.

4.2 Determinants of volume surprise induced return spillover

We estimate the parameters of the following system of equations,

$$B_{12} = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3a)$$

$$B_{14} = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3b)$$

$$B_{22} = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3c)$$

$$B_{24} = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1 \quad (3d)$$

where the dependent and independent variables are as follows.

$B_{ij} = b_{ij} * pdum$ {pdum =1, if $b_{ij} > 0$ significantly; pdum =-1, if $b_{ij} < 0$ significantly; pdum =0, if b_{ij} is neither for $i = \{1, 2\}$ and $j = \{2, 4\}$ }

$X_1 = \text{Log}(MVE)$ - MVE-ADR and MVE-home security

$X_2 = \text{Log}(TR)$ - Turnover ratio, TR is $\text{Volume}(t)/MVE(t-1)$

$X_3 = \text{Olapd}$ {1, if home trading hours partially overlap (<1/2 of trading day) with those in US; -1 if mostly overlaps (>1/2 of trading day); 0, if no overlap}

$X_4 = \text{Mktd}$ {1, if developed; 0, if emerging}

Note that in the above model, market capitalization and turnover ratio are individual security (ADR and home) specific nominal variables, while the other independent variables are market/country specific indicator variables and hence identical for each home security within a home country, region, or continent and also its corresponding ADR. We estimate the parameters of the above model for each

momentum class, positive, negative, and zero momentum.

Table 5, panel I reports the OLS parameter estimates belonging to the above equations for positive momentum trading days, which find significant negative (positive) impact of firm size on the magnitude of volume induced auto and cross correlation parameters, b_{12} (b_{22} and b_{24}); however, with respect to b_{14} the coefficient (0.000 with t-stat 0.85) associated with firm size is insignificant. In contrast, change in turnover is a significant determinant (0.017 with t-stat 18.73) only with respect to the parameter, b_{24} corresponding to cross volume*return for home shares.

On the contrary, between the two country/market specific variables, time overlap seems to impact the intercept (alpha) for all the volume induced spillover parameters, b_{12} , b_{22} , b_{14} and b_{24} while whether a security belongs to a developed vs. an emerging market impacts the intercepts for only b_{12} and b_{24} (0.155 t-stat 7.39 and -0.096 t-stat -7.26). In addition, the results also find asymmetric effect of size and turnover on the volume induced spillover parameters as evidenced by the slope parameters (betas) associated with the four interaction terms, $\log(\text{MVE}) \cdot \text{Olapd}$, $\log(\text{MVE}) \cdot \text{Mktd}$, $\log(\text{TO}) \cdot \text{Olapd}$, $\log(\text{TO}) \cdot \text{Mktd}$.

Similar asymmetric findings follow from the estimates for the negative momentum and zero momentum trading days in panels II and III respectively. Among the most striking results are the alternating signs for the same parameter between positive, negative, and zero momentum days. For example, note the estimates corresponding to $\log(\text{MVE})$ and Mktdum for ADRs and home shares from panels I, II, and III. For $\log(\text{MVE})$, the estimates corresponding to ADRs (home shares) are -0.003 (0.002), 0.004

(0.001), and 0.005 (0.009) in panels I, II, and III respectively. All estimates are significant at 5% or lower levels. Clearly, in a combined dataset of ADRs, the net effect will depend on how much the positive coefficient outweighs the negative coefficient. In comparison, home shares for the combined dataset will yield perhaps a significant positive coefficient. At the end though, ADRs and home shares may both have a small positive coefficient with respect to firm size and the difference may not be significant as we find in the empirical literature. We may draw similar conclusions from the following coefficients associated with $Mktdum$: 0.002 (-0.096), 0.037 (0.035), and -0.167 (0.023) found in panels I, II, and III respectively. In the latter case, two of those six parameters, one each from ADRs and home shares are not significant.

As we mentioned earlier, the two country/market specific indicator variables may have considerable overlaps, for example, the Asian countries and Australia do not share any trading hours overlap with the US and at the same time, with the exception of Japan and Australia belong to emerging markets. These overlaps between the indicator variables may induce correlated error terms hence we consider a simultaneous system of equations approach for estimating the parameters of equations 3a-3d. We assume correlated errors and hence use a SUR methodology to estimate unbiased and efficient GLS parameters (SUR estimates not reported).

In conclusion, our results suggest the following. First, spillover coefficients denoting continuation, reversals, and no impact including those which are induced by volume surprises are to a varying extent determined by among others, firm size and liquidity the impact of which vary depending on whether a security is from a developed

or emerging market and the extent of overlap in trading times. Further, size itself may proxy for liquidity as much it does so for information and growth and hence the size effect observed in return spillover perhaps indicates compound effects of information, liquidity, and others including corporate governance and legal origin, which are documented to be determinants of growth. Second, the time varying coefficients of firm size and liquidity as determinants of return spillovers confirm the idea of time varying risk components in Campbell (1996). Thus efficacy of asset pricing models tests may be improved by selecting a limited period rather than a long time series.

5. Testing hypotheses on volatility spillover

Volatility is not observable and hence a model is necessary for estimating volatility. We estimate volatility of each ADR and its corresponding home share as the absolute values of the respective errors from the following equations (modified version of equations 2a and 2b) and test volatility persistence and volume effect on volatility persistence and asymmetric spillover in Section 5.1-5.2 below.

$$r_{i,t}^{us} = a_1 + b_{11}r_{i,t-1}^{us} + b_{12}r_{i,t-1}^h + c_{13}x_t^{us} + e_{i,t}^{us} \dots (4a)$$

$$r_{i,t}^h = a_2 + b_{21}r_{i,t-1}^h + b_{22}r_{i,t-1}^{us} + c_{23}x_t^h + e_{i,t}^h \dots (4b)$$

$$\sigma_{i,t}^A = |e_{i,t}^{us}| \dots (4c)$$

$$\sigma_{i,t}^H = |e_{i,t}^h| \dots (4d)$$

5.1 Volatility spillover across markets, persistence and volume effect

We determine the effect of volume on reducing volatility persistence in the following VAR setup and test for the effect of volume surprise on volatility denoted by $V(t)$ and

$\sigma(t)$ respectively for each ADR and its underlying home security (We drop the i subscript denoting the i th security for convenience).

$$\sigma_t^A = a_A + \sum_{i=1}^6 a_{1i} \sigma_{t-i}^A + b_{12} V_{t-1}^A + \sum_{j=1}^6 c_{1j} \sigma_{t-j}^H + d_{14} V_{t-1}^H + \varepsilon_{At} \quad \dots (5a)$$

$$\sigma_t^H = a_H + \sum_{i=1}^6 a_{2i} \sigma_{t-i}^H + b_{22} V_{t-1}^H + \sum_{j=1}^6 c_{2j} \sigma_{t-j}^A + d_{24} V_{t-1}^A + \varepsilon_{Ht} \quad \dots (5b)$$

In equations 5a and 5b above, the parameter vectors a_{mn} and c_{mn} , where $m=\{1,2\}$ and $n=\{1,2,\dots,6\}$ signify volatility persistence with regards to auto and cross correlation or volatility spillover. We test $a_{mn} = c_{mn} = 0$ for all ADRs and their underlying securities in US and home markets respectively. In addition, in the VAR framework, rejecting hypotheses $d_{i4} = 0$ implies causality, i.e. lag cross security volatility causes volatility. Further, the model tests whether volume reduces volatility persistence.

Table 6 contains country wise breakdown of the results from the above hypotheses tests and indicate continued significant lag dependence in auto and cross volatility for both ADR and its underlying home security volatility. However, the results in Table 6, columns 3 and 5 for ADR, and columns 7 and 9 for home securities find little evidence of volatility dependence on auto or cross volume. Note the vast majority of parameter estimates, mostly 0.000 and the large number of insignificant estimates provide evidence of negligible volume effect on volatility. On the other hand, the large number of significant auto correlation coefficients suggests lag volume does not reduce volatility clustering. Our conclusion that volume surprise does not reduce volatility persistence for either ADRs or their underlying home shares raises the possibility that volume surprise may not contain ‘information’ related to price discovery but may just

reveal transient liquidity in the way of net buy/sell positions in the market.¹⁰

5.2 Asymmetric Volume Induced Volatility

In this section, we use a variation of the model in Abramov et al. (2006) and estimate the parameters of equations 6a and 6b below for each ADR and its underlying home share, where V_t is volume surprise and volatility, $\sigma(t)$ is estimated as absolute value of the error terms in equations 4a-4b.

$$\sigma_t^A = a_A + \sum_{i=1}^6 b_{1i} \sigma_{t-i}^A + c_1 e_{t-1}^A + d_1 V_{t-1}^A e_{t-1}^A + \sum_{j=1}^6 g_{1j} \sigma_{t-j}^H + k_1 e_{t-1}^H + l_1 V_{t-1}^H e_{t-1}^H + \varepsilon_{1t} \quad (6a)$$

$$\sigma_t^H = a_H + \sum_{i=1}^6 b_{2i} \sigma_{t-i}^H + c_2 e_{t-1}^H + d_2 V_{t-1}^H e_{t-1}^H + \sum_{j=1}^6 g_{2j} \sigma_{t-j}^A + k_2 e_{t-1}^A + l_2 V_{t-1}^A e_{t-1}^A + \varepsilon_{2t} \quad (6b)$$

Note that the parameter vectors b_i and g_i in equations 6a and 6b signify volatility auto and cross correlation; c_i and k_i denote leverage effect; and d_i and l_i indicate volume induced volatility spillover. While we confirmed the volatility persistence and comovement in the prior section, in this section we focus on testing the null hypotheses, $c_i = 0$, $d_i = 0$, $k_i = 0$ and $l_i = 0$ for all $i=\{1,2\}$ denoting the US and home markets respectively, which are based on the empirical implications in Wang (1993, 1994) and Campbell et al. (1993) and tested by Abramov et al. (2006) in the context of single listed US domestic securities, as we extend those to cross listed securities.

Table 7A contains country wise breakdown of the means of estimated leverage and volume induced spillover parameters, $\mu(c_i)$, $\mu(d_i)$, $\mu(k_i)$, and $\mu(l_i)$. Further, the last row contains summary statistics of those parameters along with the number of positive and negative significant parameters. The full sample means of the four volume induced

¹⁰ We observe that with the addition of volume, a few securities show the volatility auto correlations reverse to negative coefficients (2 each in Australia, Brazil, Mexico, Switzerland, and UK). It's not clear whether this is data induced spurious effect or under certain conditions volatility serial autocorrelation is indeed negative.

spillover parameters, c_i , d_i , k_i , and l_i are -0.038, -0.08, -0.04, and -0.36 respectively while the number of negative coefficients associated with parameters d_1 , e_1 , and e_2 exceeds 80 percent of the total number of observations; only d_2 seems to have a relatively higher number of positively significant coefficients (49 positive and 92 negative).

5.3 Determinants of volume induced volatility spillovers

The factors determining return spillover (Table 5) affect volatility spillover differently. Unlike its effect on volume induced return spillover, for volatility spillover, firm size is related to cross correlation, the sign is negative, and the effect is only on ADRs and not home shares. However, similar to the results on return spillover, we find the impact of size and liquidity widely vary across the three momentum classes.

We also estimate the GLS parameters using SUR methodology for a system of equations similar to 3a-3d to isolate the effect of firm size on the volume induced spillover parameter estimates. The SUR estimates indicate multidimensional asymmetric effects of firm size on volume induced spillover parameters. First, there is asymmetric size effect on the spillover parameters related to ADRs and their underlying home shares; second, there is also asymmetry in size effect on the spillover parameters for those securities with continuation vs. reversal attributes; finally, there is asymmetry between auto and cross correlation parameters with respect to size effect on the spillover parameters.

The significant coefficients 0.016 (t-stat 3.08) and -0.007 (t-stat -2.16) associated

with the interaction term, $\log(\text{MVE}) \times \text{Spild}$ in the model for the home share spillover parameters, d_2 and e_2 respectively increase (decrease) as firm size increases implying a divergence (convergence) of spillover effects for similar size firms which exhibit continuation vs. reversal. In contrast, for the sample ADRs, while there is no measurable size effect on the spillover parameters, both auto and cross serial autocorrelation effects show a significant higher (lower) mean (0.074 with t-stat 2.35 and 0.132 with t-stat 3.35 respectively) for those securities which exhibit continuation (reversal). The adjusted R^2 ranges from 51% to 64%, which is a quantum leap in model performance relative to those in return spillover.

6. Conclusion

We empirically investigate the effect of volume on return and volatility spillover for 168 ADRs and their underlying securities listed in 26 developed and emerging market countries. We find that a) cross security lag volume determines security volume in multi market trading; b) a lag interaction term residual volume*return casts asymmetric impact on return for ADRs and their underlying home securities; c) a lag interaction term residual volume*return residual casts asymmetric impact on volatility for ADRs and their underlying home securities. Our findings indicate short term return and volatility predictability in cross listed multi market trading shares and that volume induced return and volatility spillover are asymmetric on multiple dimensions. We find that there are asymmetries between ADRs and their underlying home shares, between return and volatility spillovers, and also between auto and cross correlation parameters.

Further size and liquidity asymmetrically determine the volume induced spillovers based on whether there is momentum (positive or negative feedback) trading. In certain instances, momentum intensifies or mutes the size and liquidity effect on volume induced spillover in securities from developed vs. emerging markets or from countries with low, high, or no time overlap with the US trading day. We also observe systematic differences between return and volatility spillover, which seem to support Diebold and Yilmaz (2009) conclusions from a study on international equity markets that ‘returns exhibit trend but no bust and volatility no trend but only bust’.

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Table 1: Sample breakdown into countries included in MSCI ACWI index, which is composed of 23 developed and 23 emerging markets indexes as below.

	Total	MSCI	Sample
Nos. of non-US issuers (countries) listed on NYSE Market as of Dec 2012	525 (46)		
Nos. of ADRs (countries with ≥ 1 ADR) listed on NYSE as of Dec 2012	259 (33)		
ADRs (Countries) included in Developed Markets		81 (16)	84 (17)
Australia		6	6
Belgium		2	2
Denmark		1	1
Finland		1	1
France		7	7
Germany (1 preferred)		4	3
Ireland		3	3
Israel		2	2
Italy		4	4
Japan		15	15
Netherlands (3 NY Registry Shares)		5	2
Norway		1	1
Portugal		1	1
Spain		3	3
Switzerland (1 NY Registry Shares)		5	4
UK		23	23
Luxembourg*			3
ADRs (Countries) included in Emerging markets		148 (15)	86 (12)
Brazil (1 preferred)		25	16
Chile (1 preferred)		12	10
China		60	12
Colombia		2	
Greece		1	1
India		8	6
Indonesia		1	1
Mexico (1 Global Share)		14	12
Peru		2	
Philippines		1	1
Russia		2	
South Africa		7	6
South Korea		8	7
Taiwan		5	5
Turkey		1	
Argentina*			10

Table 2: The left panel contains descriptive statistics on daily trading volume, $v_t^i = \ln(N_t^i)$ on the t th day for each of our sample ADR listed on NYSE and its underlying home share listed on the respective home country stock exchange. The right panel includes descriptive statistics on daily return computed as $r_t^i = \ln(P_t^i / P_{t-1}^i)$ where P_t^i denotes closing price of security i on the t th day, on those ADRs and their underlying shares listed on their respective home country stock exchanges. The row containing country averages are in block letters.

	ADR	Volume	Home	Volume	ADR	Return	Home	Return
	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
AWCMY	106494.6	217794.7	8629968.3	7648495.4	0.000	0.030	-0.015	0.030
BHP	1651681.8	2108531.9	12860506.1	8298379.8	0.000	0.024	-0.008	0.021
JHX	3318.2	9742.3	2084845.4	1499590.8	0.001	0.028	-0.014	0.027
OBTLY	3170.6	9041.3	112548.7	374384.9	-0.003	0.055	-0.016	0.064
SSN	729961.8	1194819.6	7683065.8	22561605.4	-0.001	0.085	-0.029	0.081
WBC	87878.7	112997.7	5885210.6	4080325.1	0.000	0.021	-0.008	0.017
Australia	430417.6	608821.3	6209357.5	7410463.6	0.000	0.041	-0.015	0.040
BUD	1147420.0	932431.6	2179175.4	1272326.2	0.001	0.016	-0.009	0.016
DEG	131390.9	118219.1	388267.7	264123.5	0.000	0.022	-0.012	0.025
Belgium	639405.5	525325.4	1283721.6	768224.8	0.000	0.019	-0.011	0.020
BBD	7831290.7	5413923.4	611502.8	2170851.2	0.001	0.030	-0.011	0.025
BSBR	6463344.1	6028380.0	2876502.8	3306821.3	-0.001	0.023	-0.016	0.023
CIG	2296613.0	1802382.3	145640.6	176815.6	0.001	0.030	-0.014	0.028
CPL	394862.3	298425.8	841843.7	1001381.7	0.000	0.024	-0.014	0.022
EBR	868216.8	649703.2	1472277.3	1187710.0	-0.001	0.032	-0.018	0.032
ERJ	769524.0	589715.1	1082098.9	1193192.7	0.000	0.028	-0.017	0.031
FBR	799096.0	854860.9	711720.5	815748.1	0.000	0.031	-0.018	0.033
GFA	2222667.2	1460041.5	6755572.5	6374278.7	-0.001	0.046	-0.027	0.048
OIBR	808798.2	1468041.4	204567.3	751249.3	0.000	0.037	-0.011	0.036
PBR	11578084.3	8741422.2	4177294.0	3311323.6	0.000	0.030	-0.015	0.029
SBS	1557926.4	1197217.6	1197656.2	873092.0	0.001	0.030	-0.017	0.029
SID	4449706.7	3792935.1	4402944.9	3101419.8	0.000	0.036	-0.018	0.036
TSU	602587.8	672115.9	1180677.4	2057492.8	0.000	0.035	-0.023	0.042
UGP	274444.1	290466.5	610171.5	645401.6	0.001	0.026	-0.011	0.024
VALE	17798648.6	12275039.0	2863466.7	1933156.4	0.001	0.030	-0.014	0.028
VIV	435350.5	779685.3	42049.6	138546.2	0.000	0.027	-0.011	0.026
Brazil	3696947.5	2894647.2	1823499.2	1814905.1	0.000	0.031	-0.016	0.031
BCA	30538.0	76470.7	261778399.9	345751579.6	0.000	0.022	-0.010	0.019
BCH	16492.1	22097.6	29323638.9	62441199.0	0.001	0.021	-0.009	0.019
BSAC	312371.0	467508.7	64027850.4	117768540.9	0.000	0.020	-0.012	0.023
CCU	115006.4	209918.6	352367.3	1472685.6	0.000	0.019	-0.010	0.021
ENI	368087.4	450920.1	22815900.3	31187131.4	0.000	0.022	-0.014	0.026
EOC	142524.2	225169.9	6411800.0	6600765.9	0.000	0.020	-0.012	0.022

	ADR	Volume	Home	Volume	ADR	Return	Home	Return
	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
LFL	364776.8	446224.1	531140.6	1124445.2	0.000	0.026	-0.013	0.030
PVD	22688.4	90378.3	348933.6	914631.9	0.000	0.018	-0.006	0.019
SQM	458118.1	865570.0	360156.7	435979.3	0.000	0.024	-0.014	0.032
VCO	25391.5	196190.0	469215.3	927888.3	0.000	0.019	-0.007	0.021
Chile	185599.4	305044.8	38641940.3	56862484.7	0.000	0.021	-0.011	0.023
NVO	707226.5	1045171.1	6854887.1	5973360.3	0.001	0.020	-0.011	0.023
Denmark	707226.5	1045171.1	6854887.1	5973360.3	0.001	0.020	-0.011	0.023
NOK	18793417.6	20495147.3	23792152.7	20913047.6	0.000	0.033	0.178	0.561
Finland	18793417.6	20495147.3	23792152.7	20913047.6	0.000	0.033	0.178	0.561
ALU	6465283.9	10537602.9	16225268.0	16759945.8	0.000	0.037	0.248	0.663
ORAN	420884.6	593845.1	8715665.8	6184855.9	0.000	0.027	0.120	0.487
SNY	1668879.6	1646706.2	3794283.0	2308165.1	0.000	0.019	-0.011	0.020
TOT	1526951.7	1511071.7	7193160.5	5065749.2	0.000	0.019	0.259	0.660
VE	219704.1	318086.7	2170266.1	1512829.4	0.000	0.025	-0.015	0.028
France	2060340.8	2921462.5	7619728.7	6366309.1	0.000	0.025	0.120	0.372
FMS	167043.4	373007.7	905551.0	723468.2	0.000	0.021	-0.013	0.024
Germany	167043.4	373007.7	905551.0	723468.2	0.000	0.021	-0.013	0.024
ACH	597633.3	720383.4	37932556.0	37154771.4	0.000	0.036	-0.019	0.039
CEA	41158.5	77681.8	19465710.4	24965697.9	0.000	0.039	-0.025	0.046
CHA	174718.7	182477.6	106509696.1	86342699.5	0.000	0.028	-0.016	0.028
GSH	38784.8	50180.7	7666724.2	8566020.7	0.000	0.029	-0.017	0.033
HNP	201151.4	190009.5	26051297.5	16947826.3	0.000	0.027	-0.016	0.029
LFC	1000097.5	1797104.9	70616901.3	63346083.6	0.001	0.028	-0.014	0.028
PTR	433895.5	456955.7	122044955.5	117598760.7	0.001	0.025	-0.013	0.027
SHI	36530.3	46800.0	21895148.9	23261070.6	0.000	0.036	-0.023	0.042
SMI	252630.4	353860.4	89293622.2	122292247.7	-0.001	0.038	-0.022	0.042
SNP	357121.0	368477.1	144423359.2	132229071.2	0.001	0.027	-0.015	0.029
YZC	261378.4	308914.1	20781747.1	15142717.3	0.000	0.038	-0.021	0.043
ZNH	58610.4	90933.6	17858700.0	16160609.0	0.000	0.039	-0.024	0.045
Hong Kong	287809.2	386981.6	57045034.9	55333964.7	0.000	0.032	-0.019	0.036
HDB	1198820.4	1229344.1	3453627.5	4552431.9	0.001	0.029	-0.015	0.025
IBN	1462057.1	1576927.0	3453846.8	5268845.1	0.000	0.037	-0.021	0.037
INFY	1930013.3	1720751.0	1388641.7	1608781.2	0.000	0.026	-0.009	0.027
RDY	355174.5	333328.2	433731.1	395660.8	0.001	0.024	-0.017	0.026
TTM	1281029.4	996650.2	15118853.1	12713395.4	0.001	0.032	-0.021	0.037
WIT	831134.6	705128.0	2727297.1	2376250.5	0.000	0.033	-0.021	0.036
India	1176371.6	1093688.1	4429332.9	4485894.2	0.000	0.030	-0.017	0.031
TLK	333484.7	427109.8	125460933.8	119351720.7	0.000	0.033	-0.015	0.033
Indonesia	333484.7	427109.8	125460933.8	119351720.7	0.000	0.033	-0.015	0.033
CRH	174665.2	264259.6	1691198.1	1247485.4	0.000	0.028	-0.017	0.031
ELN	3937993.8	7138215.6	317843.1	491379.0	0.000	0.055	-0.025	0.066
IRE	149659.6	433499.8	14882791.7	39197450.0	-0.001	0.045	-0.027	0.058

	ADR	Volume	Home	Volume	ADR	Return	Home	Return
	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
Ireland	1420772.9	2611991.6	5630611.0	13645438.1	0.000	0.043	-0.023	0.052
BSI	9422.1	17346.8	49826.6	74868.3	0.000	0.031	-0.015	0.031
TEVA	4506775.8	3853839.6	498157.2	339807.3	0.000	0.014	-0.006	0.017
Israel	2258098.9	1935593.2	273991.9	207337.8	0.000	0.023	-0.011	0.024
E	416488.1	449037.4	19537939.8	15758133.7	0.000	0.019	-0.010	0.020
LUX	140186.7	155489.3	840328.7	792943.7	0.000	0.021	-0.013	0.023
TI	234536.8	259966.2	108526218.2	75386724.6	0.000	0.024	-0.014	0.024
Italy	263737.2	288164.3	42968162.2	30645934.0	0.000	0.021	-0.012	0.022
ATE	17808.0	41038.9	2552081.8	1392821.3	0.000	0.030	-0.017	0.033
CAJ	361018.7	300616.3	5239679.1	2631220.2	0.000	0.021	-0.011	0.024
DCM	180133.2	219645.4	8808326.2	5233942.8	0.000	0.025	-0.011	0.025
HMC	378307.9	472726.0	5400265.8	3581320.3	0.000	0.021	-0.012	0.026
IX	22177.1	39522.5	8775088.8	11349957.3	0.000	0.030	-0.016	0.035
KNM	5794.8	14756.3	1067598.8	769184.9	0.000	0.026	-0.015	0.030
KUBTY	25467.9	46852.7	4490492.8	3447000.3	0.000	0.026	-0.015	0.031
KYO	43821.6	57841.3	1658150.2	1112005.6	0.000	0.024	-0.013	0.027
MFG	504123.9	623562.6	152123914.6	109808812.8	-0.001	0.029	-0.017	0.032
MTU	1679461.2	1773020.6	47726354.2	37235254.1	0.000	0.026	-0.015	0.029
NJ	103520.8	194086.5	1396761.0	882438.4	0.001	0.025	-0.013	0.027
NMR	478477.0	821076.7	23760659.9	26755745.4	0.000	0.028	-0.015	0.030
NTT	267816.0	326904.5	2104008.1	1329373.7	0.000	0.021	-0.012	0.022
PCRFY	233277.7	294537.9	7061701.1	8612922.3	0.000	0.021	-0.012	0.024
SMFG	912327.5	1619312.7	8919347.1	4517789.3	0.001	0.019	-0.010	0.021
SNE	863404.0	1226713.9	5881011.2	6828041.7	0.000	0.024	-0.013	0.027
TM	430302.8	603530.4	7906362.9	4610259.1	0.000	0.019	-0.011	0.023
Japan	382778.8	510337.9	17345400.2	13535181.7	0.000	0.024	-0.013	0.027
AMX	8979495.3	5743675.6	83661778.6	46718095.8	0.001	0.023	-0.011	0.024
ASR	107931.9	163372.1	367404.5	921356.0	0.001	0.027	-0.009	0.028
CX	6837332.3	7699999.0	23462304.4	25307200.4	0.000	0.030	-0.015	0.033
FMX	848418.6	692366.4	3374790.3	2797439.7	0.001	0.024	-0.012	0.026
GMK	24930.6	36393.3	733272.2	1393031.1	0.000	0.032	-0.013	0.033
HXM	448954.7	529627.9	1192268.8	2146708.5	-0.001	0.038	-0.019	0.046
IBA	15062.7	21256.9	104002.5	284652.4	0.000	0.031	-0.006	0.028
ICA	139225.4	261751.3	1759555.0	2696315.8	0.000	0.036	-0.018	0.037
MXT	36546.9	223948.6	869296.0	1178527.8	-0.001	0.043	-0.026	0.043
PAC	208488.2	522053.4	916828.3	2051511.9	0.000	0.023	-0.012	0.023
SIM	74588.3	154176.4	219130.6	583841.4	0.000	0.053	-0.015	0.051
Mexico	1610997.7	1458965.5	10605511.9	7825334.6	0.000	0.033	-0.014	0.034
AEG	546042.5	675418.8	8044119.2	6048141.0	0.000	0.031	0.098	0.284
ING	1414358.5	1793438.9	17220268.3	11514542.4	0.000	0.032	0.057	0.233
PHG	1345248.2	1301142.2	7930283.6	5486333.7	0.000	0.027	0.100	0.281
Netherlands	1101883.1	1256666.6	11064890.4	7683005.7	0.000	0.030	0.085	0.266

	ADR	Volume	Home	Volume	ADR	Return	Home	Return
	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
STO	999791.2	1046917.7	8354798.3	7812458.0	0.000	0.023	-0.011	0.022
Norway	999791.2	1046917.7	8354798.3	7812458.0	0.000	0.023	-0.011	0.022
PHI	187740.8	170256.7	165224.1	247417.4	0.000	0.022	-0.009	0.025
Philippines	187740.8	170256.7	165224.1	247417.4	0.000	0.022	-0.009	0.025
PT	279413.0	855945.3	4114881.4	3795258.7	0.000	0.023	-0.013	0.024
Portugal	279413.0	855945.3	4114881.4	3795258.7	0.000	0.023	-0.013	0.024
AU	1387149.4	1201112.4	805037.0	1220785.8	0.000	0.030	-0.016	0.030
DRD	76075.7	71771.2	944155.8	939174.1	0.000	0.031	-0.018	0.031
GFI	2452469.3	2869534.9	1574433.6	2181303.6	0.000	0.035	-0.018	0.035
HMY	1595742.3	1552556.5	1065287.9	1119645.6	0.000	0.038	-0.018	0.039
SPP	148816.5	213324.7	5373288.5	4444745.9	0.000	0.030	-0.010	0.068
SSL	224897.6	304103.6	1563340.4	1274130.8	0.000	0.028	-0.014	0.029
S. Africa	980858.5	1035400.5	1887590.6	1863297.6	0.000	0.032	-0.016	0.039
KB	310650.0	274664.1	2040158.6	1879930.4	0.000	0.036	-0.015	0.031
KEP	577406.4	466587.3	1482562.7	1197164.5	0.000	0.028	-0.015	0.028
LPL	1091107.2	804203.9	2439861.9	2471130.6	0.000	0.033	-0.018	0.031
PKX	458191.8	380110.3	290315.9	301869.2	0.000	0.031	-0.015	0.030
SHG	95438.4	93266.5	1735764.5	1491112.4	0.000	0.031	-0.014	0.027
SKM	1045050.9	840533.4	226005.1	228119.7	0.000	0.031	-0.016	0.033
WF	10636.5	20956.3	3620508.8	4277122.5	0.000	0.037	-0.017	0.035
S. Korea	512640.2	411474.6	1690739.6	1692349.9	0.000	0.032	-0.016	0.031
BBVA	722872.8	1243640.4	32464035.7	40258983.1	0.000	0.026	0.403	1.398
SAN	2039387.7	3916645.5	2689425.3	2315336.6	0.000	0.026	0.258	0.660
TEF	1943805.2	4119102.5	35953083.0	34019971.5	0.000	0.021	0.403	1.396
Spain	1568688.5	3093129.5	23702181.3	25531430.4	0.000	0.025	0.354	1.151
ABB	2216333.3	2356541.3	11355899.3	8806441.1	0.000	0.033	-0.016	0.041
CS	973933.0	1039321.7	7240662.7	3863792.7	0.000	0.029	-0.015	0.032
NVS	1586069.1	1621393.9	6779358.6	3684363.5	0.000	0.014	-0.008	0.015
STM	1772087.2	1149713.3	7891434.2	6692322.5	-0.001	0.030	-0.017	0.032
SYT	300217.0	303644.5	439709.0	271447.8	0.001	0.020	-0.012	0.021
Switzerland	1369727.9	1294122.9	6741412.8	4663673.5	0.000	0.025	-0.014	0.028
ASX	790151.1	887111.3	42810093.8	29386804.7	0.000	0.032	-0.017	0.031
AUO	2403933.0	1778126.0	77308646.8	54013052.7	0.000	0.032	-0.018	0.031
CHT	688174.6	737544.8	12347310.9	10330140.0	0.000	0.017	-0.006	0.013
SPIL	1403183.1	1265599.5	19086990.5	13175345.8	0.000	0.035	-0.017	0.033
TSM	9872584.2	6863761.7	67184614.2	64449361.0	0.000	0.033	-0.011	0.026
UMC	3687556.3	2518071.8	61268704.3	43932625.6	-0.001	0.036	-0.014	0.027
Taiwan	3140930.4	2341702.5	46667726.7	35881221.6	0.000	0.031	-0.014	0.027
AV	74938.1	99254.0	10448589.5	5263321.6	0.000	0.024	-0.014	0.025
AZN	976544.6	1019286.7	4584643.6	3872894.5	0.000	0.018	-0.011	0.021
BBL	774249.8	901318.5	14727426.7	10280865.3	0.001	0.028	-0.014	0.029
BCS	1211229.8	2215159.1	40801768.7	35010243.8	0.000	0.033	-0.018	0.038

	ADR	Volume	Home	Volume	ADR	Return	Home	Return
	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
BP	4845448.2	9738717.8	46219375.9	37031632.6	0.000	0.018	-0.012	0.021
BT	127544.8	184707.8	31275582.1	28920061.2	0.000	0.024	-0.015	0.027
CUK	122466.7	146596.0	1478402.5	1255890.4	0.000	0.024	-0.014	0.029
DEO	457141.0	334142.8	11324875.5	10801784.3	0.000	0.016	-0.011	0.020
GSK	1421199.8	1191401.9	12689314.1	10493142.2	0.000	0.017	-0.011	0.020
HSBC	1341768.5	1527827.5	38812387.8	25734493.6	0.000	0.018	-0.010	0.021
IHG	156098.6	268945.7	1988399.9	1562739.5	0.000	0.024	-0.013	0.025
UK	1046239.1	1602487.1	19486433.3	15475188.1	0.000	0.022	-0.013	0.025

Figure 1: XY plots of risk denoted by standard deviation against mean return

The row contains mean daily returns (X axis) and the corresponding standard deviations (Y axis) for home shares and ADRs in columns 1 and 2 respectively.

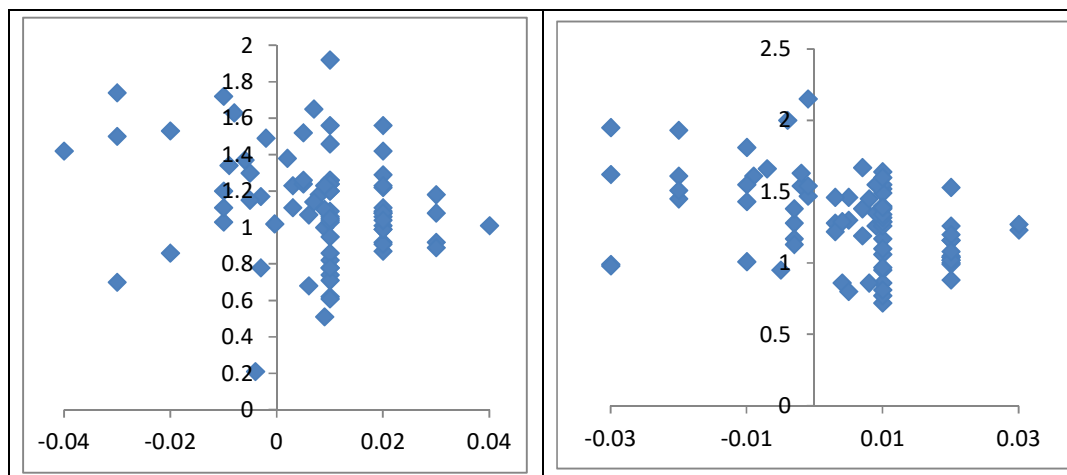


Table 3: Country wise summaries of estimated parameters of the following bivariate VAR model for ADR and home security volume

$$v_{i,t}^{us} = \alpha_1 + \beta_{11}v_{i,t-1}^{us} + \beta_{12}v_{i,t-1}^h + \phi_1v_t^{us} + V_{i,t}^{us} \dots (1a)$$

$$v_{i,t}^h = \alpha_2 + \beta_{21}v_{i,t-1}^h + \beta_{22}v_{i,t-1}^{us} + \phi_2v_t^h + V_{i,t}^h \dots (1b)$$

	ADR				Home	
Country (# of security pairs)	Mean β ; significant +; significant -	Mean φ ; significant +; significant -	Median R ² ; ADF test	Mean μ ; significant +; significant -	Mean θ ; significant +; significant -	Median R ²
Australia (6)	-.411; 0, 6	-.031; 3; 0	.19; *	-.338; 0; 6	.096; 5, 0	.30; *
Belgium (2)	-.386; 0; 2	.079; 2, 0	.17; *	-.274; 0; 2	.033; 0, 1	.40; *
Brazil (25)	-.393; 0; 24	.024; 12; 0	.19; *	-.398; 0; 24	.030; 9; 0	.22; *
Chile (10)	-.440; 0; 9	-.007; 0, 1	.20; *	-.444; 0; 9	.054; 4; 0	.20; *
Colombia (2)	-.418; 0; 2	.019; 0; 0	.21; *	-.397; 0; 2	.051; 1; 0	.15; *
Denmark (1)	-.384; 0; 1	.011; 0; 0	.16; *	-.326; 0; 1	.048; 1; 0	.27; *
Finland (1)	-.232; 0; 1	-.038; 0; 1	.17; *	-.135; 0; 1	.084; 1; 0	.73; *
France (6)	-.389; 0; 6	.040; 4; 0	.19; *	-.257; 0; 6	.040; 6; 0	.41; *
Germany (3)	-.383; 0; 3	.078; 3; 0	.19; *	-.246; 0; 3	.038; 3; 0	.43; *
Hong Kong (12)	-.398; 0; 14	-.001; 0; 5	.19; *	-.326; 0; 14	.083; 14; 0	.22; *
India (6)	-.361; 0; 6	.001; 1; 1	.17; *	-.339; 0; 6	.100; 5; 0	.26; *
Indonesia (1)	-.381; 0; 1	.029; 1; 0	.16; *	-.355; 0; 1	.103; 1; 0	.22; *
Ireland (3)	-.375; 0; 3	.022; 1; 0	.16; *	-.348; 0; 3	.058; 1; 0	.30; *
Israel (2)	-.388; 0; 2	.023; 0; 0	.20; *	-.420; 0; 2	.086; 1; 0	.22; *
Italy (5)	-.361; 0; 5	-.009; 0; 1	.17; *	-.308; 0; 5	.076; 5; 0	.26; *
Japan (17)	-.395; 0; 17	-.016; 0; 3	.18; *	-.284; 0; 17	.059; 16, 0	.30; *
Mexico (12)	-.395; 0; 11	.048; 5; 0	.18; *	-.360; 0; 11	.030; 4; 0	.29; *
Netherlands (6)	-.372; 0; 6	.022; 2; 0	.19; *	-.273; 0; 6	.067; 6; 0	.42; *
Norway (1)	-.397; 0; 1	-.041; 0; 1	.20; *	-.347; 0; 1	.056; 1; 0	.26; *
Peru (1)	-.334; 0; 1	.002; 0; 0	.22;	-.455; 0; 1	.098; 0; 0	.20;
Philippines (1)	-.398; 0; 1	-.032; 0; 1	.17; *	-.404; 0; 1	.106; 1; 0	.19; *
Portugal (1)	-.428; 0; 1	.039; 1; 0	.20; *	-.314; 0; 1	.050; 1; 0	.26; *
Russia (2)	-.328; 0; 2	-.023; 0; 1	.21;	-.456; 0; 2	.131; 1; 0	.21;
S Africa (6)	-.359; 0; 6	-.010; 0; 1	.22; *	-.378; 0; 6	.164; 5; 0	.26; *
S Korea (7)	-.365; 0; 7	-.020; 0; 1	.18; *	-.375; 0; 7	.116; 7; 0	.18; *
Spain (3)	-.383; 0; 3	.027; 1; 0	.19; *	-.272; 0; 3	.033; 3; 0	.47; *
Switzerland (5)	-.382; 0, 4	.037; 2; 0	.17; *	-.222; 0; 4	.037; 4; 0	.48; *
Taiwan (6)	-.386; 0; 7	-.020; 0; 3	.21; *	-.314; 0; 7	.113; 7; 0	.30; *
UK (11)	-.385; 0; 21	.035; 9; 0	.18; *	-.302; 0; 21	.040; 15, 0	.35; *
All						

Table 4A: Country wise summaries of select estimated parameters of the following bivariate VAR model for ADR and home security return, r_t . The select parameters are the coefficients corresponding to auto and cross lag correlation and auto and cross lag return*volume residual for each security.

$$r_{i,t}^{us} = a_1 + b_{11}r_{i,t-1}^{us} + b_{12}r_{i,t-1}^{us}V_{i,t-1}^{us} + b_{13}r_{i,t-1}^h + b_{14}r_{i,t-1}^hV_{i,t-1}^h + b_{15}x_t^{us} + e_{i,t}^{us} \dots (2a)$$

$$r_{i,t}^h = a_2 + b_{21}r_{i,t-1}^h + b_{22}r_{i,t-1}^hV_{i,t-1}^h + b_{23}r_{i,t-1}^{us} + b_{24}r_{i,t-1}^{us}V_{i,t-1}^{us} + b_{25}x_t^h + e_{i,t}^h \dots (2b)$$

	ADR			ADR	Home			Home
Country (# of security pairs)	Mean b_{11} ; # + sig # - sig	Mean b_{12} ; + sig - sig	Mean b_{13} ; + sig - sig	Mean b_{14} ; + sig - sig	Mean b_{21} ; + sig - sig	Mean b_{22} ; + sig - sig	Mean b_{23} ; + sig - sig	Mean b_{24} ; + sig - sig
Australia	-.122; 0,5	.020; 1,1	.093; 4,1	.016; 1,1	-.181; 0,6	-.005; 1,1	.183; 5,0	.029; 3,0
Belgium	-.166; 0,2	.027; 0,0	.183; 2,0	-.064; 0,0	.019; 0,0	-.049; 0,0	-.021; 0,0	.007; 0,0
Brazil	-.085; 1,13	-.004; 2,8	.152; 20,0	.006; 5,0	-.183; 0,23	.016; 8,0	.167; 23,0	.021; 6,0
Chile	-.279; 0,9	-.019; 1,3	.241; 8,1	.006; 2,0	-.214; 0,8	-.005; 0,1	.163; 7,0	.000; 1,0
Denmark	-.153; 0,1	.014; 1,0	.118; 1,0	.071; 1,0	-.047; 0,1	.017; 0,0	-.004; 0,0	.079; 0,0
Finland	-.001; 0,0	-.007; 0,0	.030; 0,0	-.001; 0,0	-.532; 0,1	-.040; 0,0	.005; 0,0	-.004; 0,0
France	-.090; 0,4	.011; 1,0	.113; 3,1	.007; 2,0	-.107; 0,4	-.004; 0,2	.132; 5,0	.027; 2,1
Germany	-.110; 0,1	.000; 1,0	.096; 3,0	.002; 0,0	-.148; 0,3	-.022; 0,1	.139; 3,0	.002; 1,0
Hong Kong	-.162; 0,12	.004; 0,2	.164; 13,0	.029; 6,0	-.479; 14,0	.017; 5,0	.555; 14,0	.003; 2,0
India	-.101; 0,6	.003; 1,0	.104; 6,0	.019; 1,0	-.194; 0,6	-.027; 0,2	.270; 6,0	.030; 4,2
Indonesia	-.139; 1,0	-.055; 0,1	.154; 1,0	.023; 0,0	-.277; 0,1	-.080; 0,1	.362; 1,0	-.030; 0,0
Ireland	-.127; 0,2	.042; 1,0	.195; 2,0	-.017; 1,0	-.268; 0,3	-.068; 0,2	.338; 3,0	.015; 2,0
Israel	-.110; 0,1	.002; 0,0	.161; 1,0	-.001; 0,0	-.226; 0,2	-.073; 0,1	.352; 2,0	.055; 1,0
Italy	-.056; 0,5	-.025; 0,1	.053; 4,0	.043; 3,0	-.182; 0,4	.045; 2,1	.167; 3,0	.009; 1,0
Japan	-.084; 0,13	.015; 2,0	.089; 13,0	.026; 8,1	-.110; 0,15	-.013; 0,4	.154; 17,0	.065; 10,0
Mexico	.013; 0,4	-.038; 3,3	.076; 4,0	-.003; 0,3	-.257; 0,8	-.003; 0,3	.292; 10,0	.009; 3,0
Netherlands	-.079; 0,5	-.014; 0,2	.084; 4,0	.073; 3,0	-.144; 0,5	.037; 3,0	.149; 5,0	-.008; 0,2
Norway	-.091; 0,1	-.011; 0,0	.088; 1,0	-.005; 0,0	-.138; 0,1	.073; 1,0	.090; 1,0	.001; 0,0
Philippines	-.030; 0,0	.006; 0,0	.128; 1,0	.043; 1,0	-.384; 0,1	-.010; 0,0	.614; 1,0	-.001; 0,0
Portugal	-.172; 0,1	.015; 0,0	.219; 1,0	-.125; 0,1	-.091; 0,1	-.065; 0,1	.102; 1,0	.045; 1,0
S Africa	-.027; 0,6	.051; 3,0	.045; 4,0	.059; 2,0	-.268; 0,5	.012; 1,0	.389; 5,1	.087; 4,0
S Korea	-.209; 0,7	.072; 4,0	.177; 6,0	.053; 3,0	-.268; 0,7	.062; 5,1	.284; 7,0	.095; 4,0
Spain	-.128; 0,3	.004; 1,0	.156; 3,0	.055; 2,0	-.027; 0,1	.041; 1,0	.039; 2,0	.040; 1,0
Switzerland	-.070; 3,0	.051; 1,0	.068; 3,0	-.001; 0,1	-.165; 0,4	-.039; 0,2	.182; 4,0	.114; 3,0
Taiwan	-.147; 0,7	.023; 4,0	.128; 6,0	.032; 2,0	-.322; 0,7	.034; 2,0	.348; 7,0	.010; 5,0
UK	-.115; 0,15	-.009; 1,4	.113; 16,0	.029; 10,0	-.095; 0,15	.005; 7,0	.114; 16,0	-.009; 1,8
All-Model	-.046; 8,70	.005; 23,25	.033; 55,11	.025; 50,8	.235; 111,4	-.012; 30,38	.244; 109,8	-.012; 18,43
All-Model4	-.068; 4,111	.004; 26,29	.032; 70,13	-3e ⁻⁵ ; 35,31	.238; 132,4	-.038; 12,76	.19; 124,11	.001; 24,22

Table 4B: Select group summaries of estimated parameters of the following bivariate VAR model for ADR and home security return, r_t

In panel A, developed and emerging market countries are classified via MSCI respective index constituents. In panel B, Asia and Australia denote C1 with no overlap with US trading day; Europe and South Africa denote C2 with partial overlap with US trading day; and C3 denotes Americas with extensive (more than 2/3rd) overlap with US trading days. In panel C, three groups are identified based on daily volume and return dynamics. M+, positive momentum denote those trading days where both excess trading volume and return are positive; M-, negative momentum occurs on those days when excess trading volume and returns are negative; M0, zero momentum days denote trading days when there is no clear positive or negative momentum i.e., excess volume times return is negative.

	ADR			ADR	Home			Home
	Mean b_{11} ;	Mean b_{12} ;	Mean b_{13} ;	Mean b_{14} ;	Mean b_{21} ;	Mean b_{22} ;	Mean b_{23} ;	Mean b_{24} ;
	$b_{11} \geq 0$	$b_{12} > 0$ $b_{12} < 0$	$b_{13} \neq 0$	$b_{14} > 0$ $b_{14} < 0$	$b_{11} \geq 0$	$b_{12} > 0$ $b_{12} < 0$	$b_{13} \neq 0$	$b_{14} > 0$ $b_{14} < 0$
A.Dev v. Emer								
μ (Dev)/ μ (Emer) t-stat: Diff. in means	-.123/-.084 0.508	.020/.005 .178	.118/.135 -0.728	.000/.029 -1.79***	-.151/-.030 3.273*	-.010/.000 -0.576	.129/.364 -4.274*	.029/.023 0.389
B.Overlap								
Diff. in means- C1 vs. C2	-.124/-.099 -1.10	.011/.010 0.06	.130/.115 0.78	.030/.002 1.77***	-.277/-.161 -2.15**	-.003/-.009 0.30	.346/.145 3.01**	.025/.031 -0.33
Diff. in means- C2 vs. C3	-.099/-.117 -0.20	.010/-.020 -2.60**	.115/.156 0.83	.002/.003 0.05	-.161/-.218 -1.42	-.009/.003 0.82	.145/.207 1.16	.031/.010 1.75
Diff. in means- C1 vs. C3	-.124/-.117 0.08	.011/-.020 -1.97***	.130/.156 0.54	.030/.003 -5.10*	-.277/-.218 1.26	-.003/.003 0.33	.346/.207 -1.93***	.025/.010 -0.98
C.Momentum								
μ_{M+} vs. μ_{M-} t-stat: $\mu_{M+} > \mu_{M-}$	-.055/-.062 0.77	.012/-.007 2.20*	.071/.031 3.05*	.024/.004 1.42***	-.070/-.095 2.87*	-.031/-.009 5.48*	.112/.091 2.69*	.007/.007 -0.06
μ_{M-} vs. μ_{M0} t-stat: $\mu_{M-} > \mu_{M0}$	-.062/-.068 0.72	-.007/-.006 -0.23	.031/.091 -5.59*	.004/.026 -2.45*	-.095/-.113 1.76***	-.009/.033 -5.72*	.091/.163 -8.42*	-.007/-.005 1.44***
μ_{M+} vs. μ_{M0} t-stat: $\mu_{M+} > \mu_{M0}$	-.055/-.068 1.52***	.012/-.006 2.27*	.071/.091 -2.00**	.024/.026 0.17	-.070/-.113 4.78*	-.031/.033 -0.18	.112/.163 -6.10*	.007/-.005 0.96

Table 5: Regression Estimates of Volume Induced Return Spillover Parameters

$$Y_i = a_0 + a_1X_1 + \dots + a_nX_n + \varepsilon_1$$

X₁= Log (MVE)

X₂= Log (Turnover ratio, Dollar volume(t)/MVE(t-1)

X₃= Olapd {1, if overlap; -1, if partial overlap; 0, if no overlap}

X₄= Mktd {1, if developed market; 0, if emerging market}

	ADR	ADR	Home	Home
I. + Momentum	Y = B₁₂	Y = B₁₄	Y = B₂₂	Y = B₂₄
C	.052 (4.09)*	.074 (3.62)*	.036 (2.61)*	.052 (5.78)*
LOG (MVE)-ADR	-.003 (-2.23)**	.000 (0.85)		
LOG (MVE)-Home			.002 (1.94)***	.008 (10.73)*
LOG(TO/10 ⁶)-ADR	-.001 (-0.54)	.002 (0.35)		
LOG(TO/10 ⁶)-Home			.000 (0.27)	.017 (18.73)*
Olapd	.150 (12.48)*	.102 (5.22)*	.087 (6.04)*	.045 (4.93)*
Mktdum	.155 (7.39)*	.002 (0.96)	-.014 (-0.70)	-.096 (-7.26)*
Olapd * LOG (MVE)	-.007 (-5.21)*	-.006 (-2.56)**	-.011 (-7.91)*	-.013 (-14.10)*
Mktd * LOG (MVE)	-.003 (-1.31)	.008 (2.28)**	.001 (0.44)	-.006 (-5.76)*
Olapd * LOG(TO/10 ⁶)	.011 (7.28)*	.009 (3.62)*	-.001 (-0.58)	-.016 (-17.80)*
Mktd * LOG(TO/10 ⁶)	.014 (5.64)*	.012 (2.87)*	.001 (2.02)**	-.031 (-20.68)*
F-stat/Adjusted R²				
II. – Momentum				
C	-.022 (-2.78)*	-.096 (-9.83)*	.031 (3.90)*	.010 (1.53)
LOG (MVE)-ADR	.004 (4.57)*	.004 (3.74)*		
LOG (MVE)-Home			.001 (2.12)**	.003 (5.91)*
LOG(TO/10 ⁶)-ADR	-.001 (-1.29)	-.013 (-10.97)*		
LOG(TO/10 ⁶)-Home			.002 (2.66)*	.003 (4.71)*
Olapd	.032 (4.24)*	-.014 (-1.47)	.017 (2.04)**	.028 (4.01)*
Mktdum	.012 (0.92)	.037 (2.29)**	.022 (1.88)***	.035 (3.62)*
Olapd * LOG (MVE)	-.005 (-5.86)*	-.002 (-1.96)***	.004 (4.73)*	-.004 (-5.34)*
Mktd * LOG (MVE)	.001 (0.57)	.003 (1.75)***	.003 (3.48)*	-.003 (-3.62)*
Olapd * LOG(TO/10 ⁶)	-.003 (-3.12)*	-.005 (-4.46)*	.007 (8.31)*	-.002 (-3.21)*
Mktd * LOG(TO/10 ⁶)	.001 (0.90)	.009 (4.62)*	.008 (6.18)*	.001 (1.26)
F-stat/Adjusted R²				
III. 0 Momentum				
C	.026 (3.02)*	.046 (4.16)*	-.066 (-7.78)*	-.009 (-0.94)
LOG (MVE)-ADR	.005 (5.67)*	.001 (0.62)		
LOG (MVE)-Home			.009 (14.18)*	.007 (8.70)*
LOG(TO/10 ⁶)-ADR	.006 (5.66)*	2.35 (1.75)***		
LOG(TO/10 ⁶)-Home			.002 (2.59)*	.004 (3.99)*
Olapd	-.023 (-2.78)**	-.004 (-0.39)	.081 (9.18)*	.063 (6.06)*
Mktdum	.014 (0.95)	-.167 (-9.07)*	.156 (12.72)*	.023 (1.57)
Olapd * LOG (MVE)	.003 (2.71)**	-.002 (-1.65)***	-.009 (-9.88)*	-.004 (-3.81)*
Mktd * LOG (MVE)	-.002 (-1.37)	.008 (4.03)*	-.011 (-10.91)*	-.001 (-1.01)
Olapd * LOG(TO/10 ⁶)	.000 (0.31)	-.000 (-0.58)	-.001 (-1.33)	.004 (4.49)*
Mktd * LOG(TO/10 ⁶)	.001 (0.38)	-.012 (-5.15)	.002 (1.22)	.001 (0.87)
F-stat/Adjusted R²				

Table 6: Country wise summary of testing the following hypotheses Granger causality via Wald test; number of rejections of null hypothesis as per significant Chi-square statistics are reported.

Null: ADR (home) security lag volatility does not Granger cause home (ADR) volatility

$$\sigma_t^A = a_A + \sum_{i=1}^6 a_{1i} \sigma_{t-i}^A + b_{12} V_{t-1}^A + \sum_{j=1}^6 c_{1j} \sigma_{t-j}^H + d_{14} V_{t-1}^H + \varepsilon_{At} \quad \dots (5a)$$

$$\sigma_t^H = a_H + \sum_{i=1}^6 a_{2i} \sigma_{t-i}^H + b_{22} V_{t-1}^H + \sum_{j=1}^6 c_{2j} \sigma_{t-j}^A + d_{24} V_{t-1}^A + \varepsilon_{Ht} \quad \dots (5b)$$

	ADR		Home		Granger causality	
Country (# of security pairs)	$B_{1m}=0$ # + sig, # - sig	$\theta_{1n}=0$ # + sig, # - sig	$\beta_{2m}=0$ # + sig, # - sig	$\Theta_{2n}=0$ # + sig, # - sig	$H_0: \beta_{2m}=0$ Home share $\sigma^2 \rightarrow$ ADR σ^2	$H_0: \theta_{1n}=0$ ADR $\sigma^2 \rightarrow$ Home share σ^2
Australia (6)	.082; 4,2	.314; 6,0	.232; 6,0	.076; 5,1	6	6
Belgium (2)	.035; 1,1	.133; 2,0	.194; 2,0	.112; 2,0	2	2
Brazil (16)	.092; 11,2	.158; 13,1	.156; 14,1	.090; 11,2	14	15
Chile (10)	.150; 10,0	.160; 9,0	.211; 10,0	.068; 9,0	9	10
Denmark (1)	.002; 0,0	.003; 1,0	.181; 1,0	.200; 1,0	1	1
Finland (1)	.042; 1,0	.000; 0,0	.002; 0,0	.378; 0,0	0	1
France (5)	.099; 3,0	.068; 2,0	.134; 3,0	.118; 2,0	2	2
Germany (1)	.001; 0,1	.176; 1,0	.302; 1,0	.003; 0,0	1	1
Hong Kong (12)	.128; 10,1	.118; 12,0	.111; 10,0	.197; 12,0	12	12
India (6)	.121; 5,0	.046; 3,0	.195; 5,0	.063; 5,0	6	6
Indonesia (1)	.166; 1,0	.105; 1,0	.166; 1,0	.055; 1,0	1	1
Ireland (3)	.060; 1,0	.071; 2,0	.075; 2,0	.123; 3,0	2	3
Israel (2)	.033; 1,0	.025; 0,0	.190; 2,0	.007; 1,0	0	2
Italy (3)	.062; 1,0	.186; 3,0	.242; 3,0	.031; 2,0	3	2
Japan (17)	.099; 14,0	.165; 17,0	.178; 17,0	.165; 16,1	16	17
Mexico (11)	.013; 4,2	.135; 6,1	.115; 6,1	.119; 6,1	8	7
Netherlands (3)	.155; 3,0	.011; 2,0	.021; 0,0	.328; 3,0	1	3
Norway (1)	.120; 1,0	.181; 1,0	.144; 1,0	.081; 1,0	1	1
Philippines (1)	.066; 1,0	.071; 1,0	.270; 1,0	.112; 1,0	1	1
Portugal (1)	.081; 1,0	.098; 1,0	.106; 1,0	.131; 1,0	1	1
S Africa (6)	.153; 6,0	.081; 5,0	.228; 6,0	.068; 5,0	5	5
S Korea (7)	.154; 7,0	.222; 7,0	.134; 6,0	.165; 7,0	7	7
Spain (3)	.193; 3,0	-.0002; 0,0	.403; 3,0	-2.33; 0,0	2	0
Switzerland (5)	.101; 3,1	.180; 3,1	.147; 4,1	.295; 5,0	5	5
Taiwan (6)	.104; 5,0	.157; 6,0	.136; 6,0	.045; 6,0	6	6
UK (11)	-.039; 5,2	.194; 9,0	.231; 10,0	.013; 7,3	9	10

Table 7A: Country wise summary of VAR parameter estimates

$$\sigma_t^A = a_A + \sum_{i=1}^6 b_{1i} \sigma_{t-i}^A + c_1 e_{t-1}^A + d_1 V_{t-1}^A e_{t-1}^A + \sum_{j=1}^6 g_{1j} \sigma_{t-j}^H + k_1 e_{t-1}^H + l_1 V_{t-1}^H e_{t-1}^H + \varepsilon_{1t} \quad (6a)$$

$$\sigma_t^H = a_H + \sum_{i=1}^6 b_{2i} \sigma_{t-i}^H + c_2 e_{t-1}^H + d_2 V_{t-1}^H e_{t-1}^H + \sum_{j=1}^6 g_{2j} \sigma_{t-j}^A + k_2 e_{t-1}^A + l_2 V_{t-1}^A e_{t-1}^A + \varepsilon_{2t} \quad (6b)$$

	ADR			ADR	Home			Home
Country	$\mu(c_1)$ # + sig, # - sig	$\mu(d_1)$ # + sig, # - sig	$\mu(k_1)$ # + sig, # - sig	$\mu(l_1)$ # + sig, # - sig	$\mu(c_2)$ # + sig, # - sig	$\mu(d_2)$ # + sig, # - sig	$\mu(k_2)$ # + sig, # - sig	$\mu(l_2)$ # + sig, # - sig
Australia	.077; 4,2	.001; 1,0	.318; 6,0	.000; 1,1	.237; 6,0	.000; 1,1	.073; 5,1	.000; 1,0
Belgium	.025; 1,1	.000; 0,0	.145; 2,0	.000; 1,0	.191; 2,0	.000; 0,0	.110; 2,0	.000; 0,0
Brazil	.092; 11,2	.000; 0,2	.163; 13,1	.000; 1,0	.163; 14,1	.000; 4,0	.086; 11,2	.000; 1,1
Chile	.150; 10,0	.000; 0,0	.160; 8,0	.000; 1,0	.222; 10,0	.000; 2,1	.066; 9,0	.000; 0,0
Denmark	.002; 0,0	.000; 0,0	.003; 1,0	.000; 1,0	.182; 1,0	.000; 1,0	.201; 1,0	.000; 0,0
Finland	.043; 1,0	.000; 0,0	-.0001; 0,0	.000; 0,0	.002; 0,0	-.005; 0,1	.376; 0,0	.000; 0,0
France	.098; 3,0	.000; 0,0	.070; 2,0	.000; 1,0	.136; 3,0	.000; 2,0	.057; 2,0	.000; 0,0
Germany	-.078; 0,1	.000; 0,0	.177; 1,0	.000; 1,0	.299; 1,0	.000; 1,0	-.013; 0,0	.000; 0,0
Hong Kong	.130; 10,1	.000; 0,1	.116; 12,0	.000; 0,1	.118; 10,0	.000; 3,0	.197; 12,0	.000; 0,0
India	.075; 5,1	.000; 1,1	.078; 4,0	.000; 3,0	.242; 6,0	.000; 2,1	-.007; 5,1	.000; 2,0
Indonesia	.164; 1,0	.000; 0,0	.107; 1,0	.000; 0,0	.167; 1,0	.000; 1,0	.054; 1,0	.000; 0,0
Ireland	.067; 1,0	.000; 0,0	.074; 1,0	.000; 0,0	.078; 2,0	.000; 0,0	.130; 3,0	.000; 0,0
Israel	.038; 1,0	.000; 0,0	.031; 0,0	.000; 0,1	.211; 2,0	.000; 1,0	.008; 1,0	.000; 0,1
Italy	.067; 0,0	.000; 0,0	.183; 3,0	.000; 0,1	.233; 3,0	.000; 1,0	.036; 2,0	.000; 0,1
Japan	.100; 14,0	.000; 2,1	.168; 17,0	.000; 0,3	.171; 17,0	.000; 10,0	.171; 16,1	.000; 2,0
Mexico	.002; 4,2	.000; 0,0	.139; 8,1	.000; 2,0	.124; 6,1	.000; 1,0	.124; 7,1	.000; 1,0
Netherlands	.156; 3,0	.000; 0,0	.010; 2,0	.000; 0,0	.020; 0,0	.000; 0,0	.324; 3,0	.000; 0,0
Norway	.118; 1,0	.000; 0,0	.189; 1,0	.000; 0,0	.143; 1,0	.000; 1,0	.080; 1,0	.000; 0,0
Philippines	.067; 1,0	.000; 0,0	.070; 1,0	.000; 0,0	.276; 1,0	.000; 1,0	.120; 1,0	.000; 0,0
Portugal	.082; 1,0	.000; 0,0	.099; 1,0	.000; 0,0	.106; 1,0	.000; 1,0	.131; 1,0	.000; 1,0
S Africa	.159; 6,0	.000; 0,1	.082; 5,0	.000; 0,1	.226; 6,0	.000; 2,1	.074; 4,0	.000; 1,1
S Korea	.153; 7,0	.000; 0,3	.232; 7,0	.000; 0,2	.138; 6,0	.000; 5,0	.161; 7,0	.000; 1,1
Spain	.194; 3,0	.000; 0,0	-.0001; 0,0	.000; 0,0	.404; 3,0	-.050; 0,1	-.973; 0,0	.000; 0,0
Switzerland	.097; 3,2	.000; 0,0	.191; 4,1	.000; 0,0	.135; 4,1	.000; 0,0	.310; 5,0	.000; 3,0
Taiwan	.109; 5,0	.000; 0,3	.158; 5,0	.000; 1,0	.135; 6,0	.000; 3,0	.046; 5,0	.000; 0,2
UK	-.049; 5,2	.000; 2,2	.205; 9,0	.000; 0,2	.242; 10,0	.000; 5,1	.008; 7,2	.000; 2,1
All	.089;	.000;	.146;	.000;	.175;	-.001;	.089;	.000;

Table 7B: Select group summaries of estimated parameters of the following bivariate VAR model for ADR and home security return, r_t

In panel A, developed and emerging market countries are classified via MSCI respective index constituents. In panel B, Asia and Australia denote C1 with no overlap with US trading day; Europe and South Africa denote C2 with partial overlap with US trading day; and C3 denotes Americas with extensive (more than 2/3rd) overlap with US trading days. In panel C, three groups are identified based on daily volume and return dynamics. M+, positive momentum denote those trading days where both excess trading volume and return are positive; M-, negative momentum occurs on those days when excess trading volume and returns are negative; M0, zero momentum days denote trading days when there is no clear positive or negative momentum i.e., excess volume times return is negative.

	ADR			ADR	Home			Home
	Mean b_{11} ;	Mean b_{12} ;	Mean b_{13} ;	Mean b_{14} ;	Mean b_{21} ;	Mean b_{22} ;	Mean b_{23} ;	Mean b_{24} ;
	$c_{11} \geq 0$	$k_{12} > 0$	$d_{13} \neq 0$	$l_{14} > 0$	$c_{11} \geq 0$	$k_{12} > 0$	$d_{13} \neq 0$	$l_{14} > 0$
A.Dev v. Emer								
μ (Dev)/ μ (Emer) t-stat: Diff. in means								
B.Overlap								
Diff. in means- C1 vs. C2								
Diff. in means- C2 vs. C3								
Diff. in means- C1 vs. C3								
C.Momentum								
μ_{M+} vs. μ_{M-} t-stat: $\mu_{M+} > \mu_{M-}$	12.852/- 10.843 12.75*	0.422/0.504 -1.64	-3.527/ 0.437 -6.50*	0.492/0.452 0.89	.071/.031 3.05*	.024/.004 1.42***	-.070/-.095 2.87*	-.031/-.009 5.48*
μ_{M-} vs. μ_{M0} t-stat: $\mu_{M-} > \mu_{M0}$	-10.843/ 0.000 -8.40*	0.504/0.364 2.06**	0.437 /-1.316 3.06*	0.452/0.496 -1.06	.031/.091 -5.59*	.004/.026 -2.45*	-.095/-.113 1.76***	-.009/.033 -5.72*
μ_{M+} vs. μ_{M0} t-stat: $\mu_{M+} > \mu_{M0}$	12.852/ 0.000 11.28*	0.422/0.364 1.22	-3.527/ -1.316 -4.18*	0.492/0.496 -0.097	.071/.091 -2.00**	.024/.026 0.17	-.070/-.113 4.78*	-.031/.033 -0.18